

NANDINI SAXENA

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EDUCATION

Bharati Vidyapeeth (DU), College of Engineering, Pune

Jul 2023 – May 2027

BTech, Computer Science & Engineering — CGPA 9.3/10

Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Agentic AI, Business Intelligence, Cloud Computing

EXPERIENCE

Machine Learning Intern — DIAT, DRDO Pune

Jan 2026 – Apr 2026

- Built a transformer-based sequence model for UAV trajectory reconstruction during GPS outages using IMU sensor data, reducing trajectory RMSE by ~28% versus classical dead-reckoning integration
- Designed the dead-reckoning pipeline end-to-end — noise filtering, bias correction, and drift modelling for inertial-only navigation under GPS denial

Vice President — Google Developer Groups (GDG) BVP Pune

Jun 2025 – Present

- Lead a 12-person core team running technical programming for a 1500+ member developer community
- Secured and manage 5+ industry partnerships for events, sponsorships, and speaker engagements

Software Development Engineer Intern — MCarbon Tech

Jun 2025 – Jul 2025

- Built a smart ticketing system on Spring Boot, Flask, and Vue.js, covering backend services, API integration, and frontend delivery

SELECTED PROJECTS (GITHUB.COM/NANDINI1612)

1. [GateIO](#)

- a. Built a yaw-rate-gated deep learning system (PyTorch; LSTM, TCN+attention) predicting UAV velocity through GPS outages, cutting mean position drift to 11.2 m over 10 s outages — 7.5x better than a tuned Extended Kalman Filter (MARS-LVIG dataset)
- b. Diagnosed a data-leakage flaw in standard within-flight evaluation splits (~3x inflated accuracy) and introduced a leakage-free leave-one-flight-out cross-validation protocol

2. [Fraud Detection](#)

- a. Deployed an end-to-end fraud detection service (LightGBM, FastAPI) on 590K IEEE-CIS transactions with calibrated probabilities, per-prediction SHAP explanations, and a 32-test suite; Dockerized with Prometheus metrics and PSI-based drift monitoring
- b. Ran a temporally-split ablation across 4 model families and 3 fraud datasets, showing velocity features top SHAP importance yet add no significant predictive value once graph features are present (Wilcoxon $p = 0.23, 0.95$) — while graph features are the real driver (AUPRC +0.031, $p = 0.002$)

3. [CashPulse](#)

- a. Built a cash-flow analytics pipeline with parallel ETL (SQLite) and ELT (Postgres + dbt) tiers, orchestrated with Airflow and provisioned with Terraform
- b. Forecasts cash flow and flags anomalies (gradient-boosted regressor, Isolation Forest), gated by 54 automated tests across 2 CI pipelines; deployed as an authenticated Streamlit dashboard

WORKING PAPERS

- [GateIO](#): Yaw-rated-gated Casual Deep Learning for UAV GPS-Outage Bridging - preprint on Zenodo (DOI 10.5281/zenodo.22853978); preparing submission to IEEE/ION PLANS 2027
- **Importance Is Not Contribution**: Velocity Features Are Redundant on Real Fraud Data but Rewarded by Simulated Benchmarks – preprint on Zenodo (DOI 10.5281/zenodo.22892826)
- Disentangling Logical Validity from Statistical Utility in Synthetic Tabular Data - manuscript in preparation

SKILLS

Languages: Python, Java, SQL, JavaScript, C, TypeScript

ML / DL: PyTorch, LightGBM, XGBoost, scikit-learn, HuggingFace Transformers

Focus areas: time-series forecasting, sequence modeling (LSTM, TCN+attention), model interpretability (SHAP), experiment design (temporal cross-validation, ablation studies, statistical significance testing)

Backend / Infra: FastAPI, Flask, Spring Boot, PostgreSQL, Docker, Apache Airflow, Terraform, AWS

LEADERSHIP & ACTIVITIES

- Founder, OpporYouNity (youth opportunity & mental-health platform) · Head of PR, Null Student Chapter · Classical Bharatnatyam dancer, 10+ years (RAQS College Dance Team)